



THE UNIVERSITY OF
WESTERN
AUSTRALIA

DESK No.

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FAMILY NAME: _____

GIVEN NAMES: _____

SIGNATURE: _____

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Semester 2, 2025 EXAMINATIONS

School of Physics, Mathematics and Computing

CITS3501

Software Testing and Quality Assurance

Examination Duration: 2 hours

This is an examination WITH permitted materials

Provided by the University

1 x 10 Page Answer Booklet

Supplied by the Student

1 x double-sided A4 page of notes (printed or handwritten)

Calculator

No calculators are permitted

Instructions to Students

This exam contains 4 questions for 100 marks.

Each question is worth 25 marks.

The marks for each sub-question are indicated on the exam paper.

Candidates should attempt all questions.

Start your answer to each question on a new page of your answer book.

For all questions, be sure to explain the reasoning behind your answers. If you need to make any reasonable assumptions to answer a question, explain what they are.

Exam question papers and candidates' notes are to be collected with the examination answer booklets.

Examination candidates may only bring authorised materials into the examination room. If you are found with unauthorised material, disciplinary action will be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found. Any candidate who has brought unauthorised material into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification. Question papers and answer booklets must not be removed from the examination room.

Examination Cover Sheet

CITS5501 sample questions

Question 1

You are asked to develop tests for a Java method, `stringSimilarity()`.

The signature and API documentation are as follows:

```
1  /** Return a measure of the similarity between two strings.
2   *
3   * Strings that are identical will return 1.0; other
4   * strings will return a number greater than or equal
5   * to 0.0, but less than 1.0, according to
6   * Formula 1.1 specified in Lowenberg et al (1956).
7   *
8   * @param   s1 A string argument
9   * @param   s2 A second string to compare against s1
10  * @return  A number representing the similarity measure
11  */
12 public static double stringSimilarity(String s1, String s2)
```

Answer the following questions about the `stringSimilarity` method:

- Should the documentation for the method mention what happens when one or both of `s1` and `s2` are `null`? Why or why not? Justify your answer. (5 marks)
- Is it a precondition of the method that `s1` and `s2` must be `Strings`? Should this be mentioned explicitly in the documentation? Explain why or why not, justifying your answer. (5 marks)
- Give examples of a *failure* and a *fault* that one might find when testing the `stringSimilarity` method. Justify your answer, and explain why the two are different. (10 marks)

Question 2

You have been asked to apply the *Input Space Partitioning* technique in order to develop unit tests for the `stringSimilarity` method.

- Explain briefly the steps you would take to apply the Input Space Partitioning technique. (5 marks)
- Work through those steps in order to derive at least two test cases for the `stringSimilarity` method. Explain whether any other parameters besides `s1` and `s2` need be considered. You need not exhaustively list every characteristic and partition you would use in the process, but should mention at least two characteristics. (10 marks)
- Based on your answer to (b), describe two test cases for the `stringSimilarity` method. If any *test doubles* might be needed, state what sort. (5 marks)

Clearly explain any assumptions you need to make when answering. If there is information you would need which is not given in the provided scenario, explain what it is.